

POKHARA UNIVERSITY

Level: Bachelor Semester: Spring Year : 2019
 Programme: BE Full Marks: 100
 Course: Microprocessor and Assembly Language Programming Pass Marks: 45
 Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define Op-code and operand with Example. Differentiate between microprocessor and microcontroller. 7
 b) What do you mean by instruction cycle? Draw the timing diagram of instruction MVI M,20H. 2+6
2. a) Explain the instruction set of 8085 microprocessor on the basis of its operation. 7
 b) WAP to find the sum of given series $1+2^2+3^2+\dots+10^2$. Store the results in memory address 2040H and 2041H. 8
3. a) Identify the size, addressing modes, T-state and function of following instructions. 7
 - i. MVI A,20H
 - ii. LDA 1234H
 - iii. CMP M
 - iv. RRC
 - v. ANA B
 - vi. ADI 0FH
 - vii. LXI H,1234H
- b) What is a macro and how can it be implemented for assembly language programming? Explain a program using macros. 8
4. a) Write an assembly language program for 8086 to find the sum of two numbers which is input by the user through the key board and display the sum on the screen. 7
 b) Design an address Decoding circuit to interface an input device with eight input switches and eight LEDs output device at address 50H and 51H respectively. 8
5. a) Explain the interrupt processing in detail. Describe the methods of handling multiple interrupts. 7
 b) What is IVT? Briefly describe the conditions which cause the 8086 to perform each of the following types of interrupts: Type 0, Type 1, 8

- Type2, Type 3 and Type 4.
6. a) Describe about the pin configuration of 8237 controller and explain the types of modes available in it. 8
 b) Draw a labelled diagram of 8254 PIT controller and write down the instructions to generate 5KHz square wave for 2854 PIT. 7
 7. Write short notes on: (**Any two**) 2×5
 - a) Application of PIC.
 - b) One pass and Two pass Assemblers
 - c) ALP Development tools